

Structure in the Recurrence

Conditioning Genomic Foundation Model Pretraining on 3D Chromatin Contacts

Project technical record

Compiled August 9, 2026

No experimental results appear in this document. No model has been trained. The abstract is a proposal abstract. Every number reported comes from a data-processing, parameter-accounting or unit-test script in the repository, never from a training run. The working rule for this project is that no figure reaches a draft unless it exists first in a logged output file.

Abstract

Genomic foundation models are pretrained on DNA sequence alone, yet much of the regulatory logic they are asked to predict is carried by the genome’s three-dimensional folding, which determines which enhancers can reach which promoters. Existing work combining sequence with chromatin contact data does so after the sequence representation is already fixed: CHROME trains an encoder on individual loci and only then adds a graph-attention layer over Hi-C contacts, and Evo2HiC distills a frozen Evo 2 into a compact encoder using Hi-C as a contrastive alignment target. In both, contact data shapes which learned features are combined, never which features are learned. We propose instead to place chromatin structure inside the pretraining computation. Contact-derived features modulate the state-transition dynamics of a bidirectional Mamba backbone, so Hi-C signal changes how far information propagates along the sequence rather than only what each position encodes. The mechanism adds 1.70% to parameter count over a matched sequence-only baseline. We evaluate whether a structure-shaped representation transfers—a question no sequence-to-structure model has been asked, as a forward-citation sweep over 318 papers citing Akita and Orca confirms—using long-range regulatory and variant-effect tasks together with a perturbation-sensitivity protocol on which Evo 2 has been reported to fail. Controls include structure shuffled at pretraining time and a distance-matched rewiring that tests whether any apparent structural signal is genomic distance in disguise.

Keywords: genomic foundation models; 3D genome organisation; Hi-C; topologically associating domains; state-space models; Mamba; self-supervised pretraining; CTCF; variant effect prediction; representation transfer.

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1 The problem

An enhancer can sit 10^5 to 10^6 base pairs from the promoter it regulates. Nothing in the linear arrangement of the sequence explains how one reaches the other. The genome resolves this by folding: a ring-shaped cohesin complex extrudes a loop of DNA until it stalls at a pair of CTCF binding sites, and the resulting loop places the enhancer physically against the promoter.

The stalling condition matters for what follows. Cohesin halts only when the two CTCF motifs are *convergent*—pointing toward each other. Motif presence is insufficient; orientation decides. A sequence-only model sees the motif but receives no training signal that makes orientation pairing across a 500 kb gap consequential.

Three structures recur in the analysis. A *topologically associating domain* (TAD) is a region that contacts itself far more than its surroundings; enhancers act almost exclusively within their own TAD, so boundaries determine reachability. A *loop* is a point-to-point contact between two anchors. An *A/B compartment* is a coarse partition into open, active and closed, inactive regions.

2 Related work and the gap

Work	Seq.	Struct.	Structure enters at	Needed at inference
HiCFoundation (2026)	no	yes	pretraining	yes
Hi-Cformer (2025)	no	yes	pretraining	yes
Evo 2 (2026)	yes	no	—	no
Caduceus (2024)	yes	no	—	no
GraphReg (2022)	feat.	yes	supervised, downstream	yes
CHROME (2026)	yes	yes	stage 2, post-hoc	yes
Akita / Orca	yes	target	supervised target	no
Evo2HiC (2025)	yes	yes	contrastive distillation	no
This project	yes	yes	self-supervised pretraining	target: no

Table 1: Where 3D structure first influences the model. The bottom row is unoccupied in the current literature.

2.1 The two nearest competitors

CHROME (Ye et al., *Briefings in Bioinformatics* 27(4):bbag360, 2026) filters Hi-C against a self-avoiding polymer null model simulated over 5×10^5 chains, retaining 1.8–6% of contacts, then runs two graph-attention layers over signal-centred subgraphs with a 4 Mb receptive field at 5 kb bins. Training has two stages: the encoder is trained on centre nodes alone until validation saturates, and only afterwards are early layers frozen and the graph module attached. Supervision is 751 ENCODE assays. Structure never influences what the sequence encoder learns, and the contact graph is required at inference.

Evo2HiC (Fang et al., bioRxiv 10.1101/2025.11.18.689171) distils a frozen Evo 2 (7B) into a 3.6M-parameter CNN using two SigLIP contrastive objectives, the second aligning the student to Hi-C patch embeddings. Structure genuinely shapes a sequence encoder, but by aligning output embeddings, with no objective over nucleotides. A keyword count over the full preprint returns zero occurrences of *ClinVar*, *pathogenic*, *eQTL*, *variant*, *masked*, *MLM*, *Mamba* and *state space*.

Their third stated limitation is that they froze Evo 2 because fine-tuning it is computationally hard, and that they intend to develop strategies for adapting it with Hi-C data—this project’s question, named as work they have not done.

2.2 Two external findings that shaped the plan

Lee (arXiv:2604.07196, April 2026) probed Evo 2 (7B) on 231 TAD boundary regions and 120 convergent CTCF loops. Boundary deletions were penalised *less* than GC- and size-matched random controls (paired Wilcoxon $p = 0.405$); CTCF inversions and deletions likewise ($p = 0.021$, $p = 0.006$, in the wrong direction). Generated loops reached median enrichment 0.054 against a reference of 0.388. The paper concludes Evo 2 “has learned local CTCF grammar but misses higher-order 3D organization” and prescribes bidirectional architectures with explicit 3D contact inputs, citing Caduceus. This substantially defuses failure mode F3 (Section 6): if a 7B-parameter model with a 1M-token context cannot infer TAD-scale structure from sequence, a 7.7M-parameter model will not either.

DNALongBench (*Nature Communications* 16(1):10108, 2025) provides five long-range tasks including enhancer–target interaction, eQTL and contact-map prediction, already benchmarking Caduceus-Ph and Akita. Adopting it for evaluation removes the objection that tasks were chosen to flatter. Calibration: contact-map prediction is the hardest task in the suite, best reported correlation 0.233, by Akita.

2.3 The negative result the contribution rests on

A forward-citation sweep over 318 papers—118 citing Orca, 200 citing Akita—found no work that takes either model’s learned representations and evaluates them on a task other than predicting folding. Their variant applications all run the model twice, on reference and mutant sequence, and compare predicted contact maps. Internal embeddings are never extracted, frozen and probed, or fine-tuned into another head.

3 Method

3.1 Baseline recurrence

Per layer and direction, with channel index i and state index n :

$$(z, v) = \text{split}(W_{\text{in}}u_t) \quad (1)$$

$$x_t = \text{SiLU}(\text{Conv1d}(v)_t) \quad (2)$$

$$(\delta'_t, B_t, C_t) = \text{split}(W_x x_t) \quad (3)$$

$$\Delta_t = \text{softplus}(W_\delta \delta'_t + b_\delta) > 0 \quad (4)$$

$$A_{i,n} = -\exp(A_{i,n}^{\text{log}}) < 0 \quad (5)$$

$$\bar{A}_{t,i,n} = \exp(\Delta_{t,i} A_{i,n}) \in (0, 1) \quad (6)$$

$$h_{t,i,n} = \bar{A}_{t,i,n} h_{t-1,i,n} + \Delta_{t,i} B_{t,n} x_{t,i} \quad (7)$$

$$y_{t,i} = \sum_n C_{t,n} h_{t,i,n} + D_i x_{t,i} \quad (8)$$

3.2 The structural modification

Let $s_t \in \mathbb{R}^{d_s}$ be the structural context at position t , derived from Hi-C. Two terms are added:

$$\Delta_t = \text{softplus}(W_\delta \delta'_t + b_\delta + \textcolor{red}{W}_{\Delta s} s_t) \quad (9)$$

$$p_t = \text{softplus}(w_g^\top s_t + b_g) \geq 0 \quad (10)$$

$$\bar{A}_{t,i,n} = \exp(\Delta_{t,i} A_{i,n} - \textcolor{red}{p}_t) \quad (11)$$

The first retunes the memory length of each channel. The second is a non-negative damping that only shortens memory: at a TAD boundary the model can raise it and cut the state.

Why this is not an input feature. Appending s_t to the input would reach only $\bar{B}_t x_t$, the term governing how much of the current token enters. It cannot alter $\bar{A}_t$, which governs what survives. Because Δ enters $\bar{A}$ inside an exponential, structure here sets the effective memory horizon

$$\tau_{t,i} = \frac{-1}{\Delta_{t,i} A_{i,n} - p_t} \quad (\text{tokens}), \quad (12)$$

that is, how *far* information travels rather than what is encoded. This is the class of constraint that contrastive alignment on output embeddings cannot express, and it is the entire novelty claim.

Relationship to hard state resets. As $p_t \rightarrow \infty$, $\bar{A}_t \rightarrow 0$ and $h_t = \bar{B}_t x_t$: a hard reset. The mechanism is the soft, learned relaxation of TAD-conditioned state resetting, which is recorded as future work.

Initialisation. $W_{\Delta s} = 0$, $w_g = 0$, $b_g = -4$. The model is then numerically indistinguishable from the baseline at step zero, so any divergence is attributable to learned structural signal. Zero initialisation of $W_{\Delta s}$ does not block learning, since $\partial L / \partial W_{\Delta s} = (\partial L / \partial \Delta) s_t^\top$.

3.3 Parameter accounting

Computed by instantiating both models and summing parameter tensors (`scripts/param_accounting.py`, `scripts/test_model.py`).

Configuration	Total	Added	Δ
Sequence-only baseline	7,725,312	—	—
$d_s = 2$, with encoder	7,758,354	33,042	+0.428%
$d_s = 8$, with encoder	7,856,952	131,640	+1.704%
$d_s = 8$, no encoder (recommended)	7,856,672	131,360	+1.700%

Table 2: All configurations sit inside the 5% matched-parameter budget.

4 Implementation

4.1 Scan backends

The selective scan has two implementations, selected by device.

The reference implementation is a sequential loop over positions: correct, differentiable, and what the CPU unit tests run at small L . It is not viable at $L = 32,768$, which would require $32,768 \times 16 \times 2$ Python iterations per forward pass.

Production requires a fused kernel, and the two structural terms differ sharply in how they cooperate with the stock one. **The Δ bias is free:** `mamba_ssm`’s `selective_scan_fn` takes `delta` as an argument, so the bias is added before the call. **The permeability term p_t is not expressible** through that interface: it needs $\bar{A} = \exp(\Delta A - p)$, and folding p into Δ would require $\Delta' = \Delta + \log(g)/A_n$, which depends on the state index n . A single per-channel Δ cannot produce a decay uniform across n .

A custom Triton kernel was therefore written (`src/chromfm/scan_triton.py`), carrying p through both forward and backward. It uses one program per (batch, channel) pair, checkpoints

the state at chunk boundaries, and in the backward replays each chunk from its checkpoint before running the adjoint recurrence, keeping memory at $O(BD \frac{L}{\text{chunk}} N)$ rather than $O(BDLN)$.

The Triton kernel has never been executed. It was written on a CPU-only machine with no CUDA device and no Triton install. The recurrence and its adjoint were derived by hand and the code is syntactically complete, but nothing is numerically confirmed. `scripts/validate_kernel.py` must be run on the GPU box before any training. A scan with a subtly wrong gradient produces a smooth, plausible loss curve while training the wrong model.

Both arms must use the same backend, or the matched-compute comparison is confounded. The baseline therefore runs on this scan too, even though it never supplies p .

4.2 Dataset layer

Window		32,768 bp
Train stride		16,384 (50% overlap)
Evaluation stride		32,768 (no overlap)
Train windows		5,422
Validation windows	273	(chr9:120,029,184–128,974,848)
Test windows	242	(chr9:130,023,424–137,953,280)
Dropped	1,054 on N fraction, 799 on structural coverage, 126 buffer	

Table 3: Window index. Verified: zero training windows touch either held-out region; both evaluation splits are fully contained with no internal overlap.

Window length follows from the memory-horizon measurement rather than convention: the 16-layer relayed horizon is roughly 16,000 tokens, so 32,768 already exceeds what the stack can span.

Three policies are implemented in the dataset builder rather than in either training script, so the two arms cannot diverge on them:

- (i) **ϕ is interpolated, not stepped.** Linear interpolation between bin centres. Measured inside one window, a step function yields 7 distinct values per feature; interpolation yields 32,757. Seven values across a window is the precondition for failure mode F4.
- (ii) **N has its own token and a budget.** 12.00% of chr9 is unresolved; windows above 10% N are dropped.
- (iii) **Invalid bins are masked, never NaN.** A position is structurally valid only when both flanking bins are usable. Invalid positions receive $\phi = 0$, the standardised mean, plus a validity flag. A NaN reaching $W_{\Delta s} s_t$ would poison the entire recurrence.

Reverse-complement augmentation reverses and complements the tokens, then reverses ϕ and negates its two antisymmetric coordinates (`directionality_2Mb`, `upstream_mass_frac`).

5 Pilot data

5.1 Provenance

Source is 4D Nucleome experiment set **4DNES3JX38V5**: in situ Hi-C on GM12878, MboI with bio-dATP, Lieberman Aiden lab, published as Rao et al. 2014 (PMID 25497547), aligned to GRCh38. The multi-resolution contact matrix (4DNFIXP4QG5B) is 27.4 GB and is never downloaded; the pipeline reads a near-diagonal band directly over HTTP range requests in six and a half minutes. Boundary calls, insulation and compartment tracks are mirrored as independent validation targets, not as model inputs. Sequence is Ensembl release 113; annotation is GENCODE release 47.

5.2 Measured properties

chr9 length, from the contact matrix	138,394,717 bp
chr9 length, from the Ensembl FASTA	138,394,717 bp
Bins at 5 kb	27,679
Band entries retained	7,108,483
Band occupancy	0.6451
Bins with no balancing weight	19.31%
Unresolved sequence (N)	12.00%
Bins with a complete feature vector	21,519 (77.74%)

The two identical lengths are the first coordinate-alignment check. A second, stronger one emerged unplanned: the pipeline returned nothing for chr9 between 48 and 58 Mb. That is the centromere and the 9q12 heterochromatic block, which GRCh38 represents as a gap. The sequence file is 100% unresolved from approximately 46 to 60 Mb, and the contact matrix has no usable bins from 43.27 to 60.52 Mb. Two files from different institutions placing the same feature at the same coordinates is stronger evidence than a matching chromosome length.

5.3 Structural features

Index	Feature	Window	Under reverse complement
0–2	insulation score	100, 250, 500 kb	symmetric
3	directionality index	2 Mb	antisymmetric
4	log contact density	2 Mb	symmetric
5	upstream mass fraction	2 Mb	antisymmetric
6	short-to-long range ratio	100 kb / 2 Mb	symmetric
7	A/B compartment eigenvector	250 kb, GC-oriented	symmetric

Table 4: The symmetry column is load-bearing: failure mode F7 is the case where features 3 and 5 cancel between the forward and reverse passes.

5.4 Validation against independent tracks

Our feature	Against 4DN’s own track	Pearson r
Insulation, 100 kb window	4DNFIBMOGOZC	+0.9969
Insulation, 250 kb window	4DNFIBMOGOZC	+0.7802
Insulation, 500 kb window	4DNFIBMOGOZC	+0.4895
A/B compartment eigenvector	4DNFILYQ1PAY	+0.9759

Table 5: The 100 kb window reproduces 4DN’s track almost exactly, identifying their diamond window and confirming the implementation. Declining correlation at larger windows is the expected consequence of comparing different window sizes against a fixed reference.

A TAD and a loop are both visible in the processed data. The loop lies at chr9:136,485,000 paired with 136,625,000, 140 kb apart, at 7.5 times its distance-matched expectation, with six CTCF ChIA-PET read pairs from an independent assay at the same anchors. Its upstream anchor falls on *NOTCH1* (chr9:136,494,433–136,546,048).

Phase 1 step 3 — pilot pipeline visual validation (GM12878 in situ Hi-C, Rao et al. 2014, via 4DN, GRCh38)

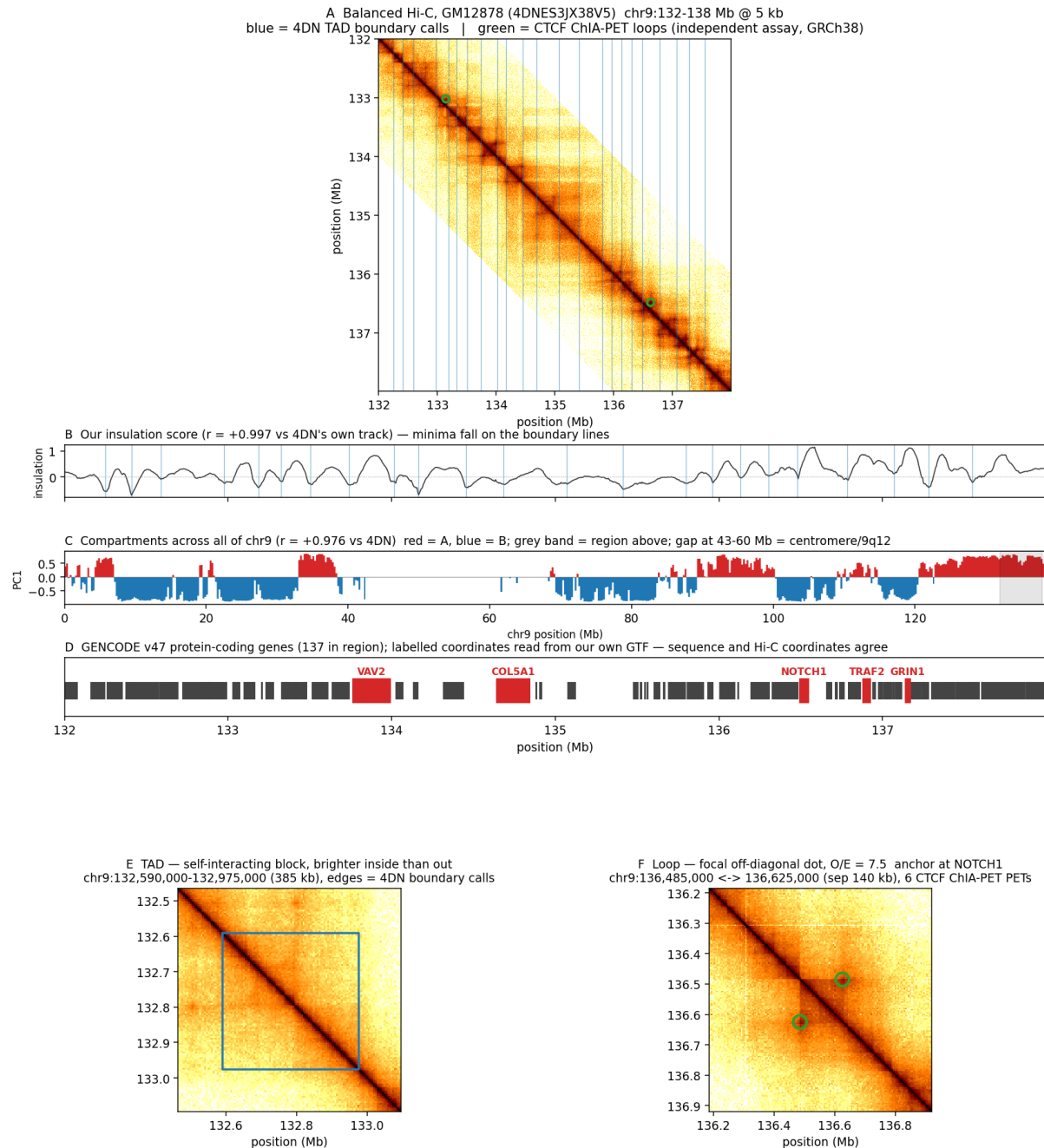


Figure 1: Pilot pipeline validation. (A) Balanced contacts with 4DN boundary calls in blue and CTCF ChIA-PET loops in green. (B) Our insulation score. (C) Compartments across all of chr9; the blank interval at 43–60 Mb is the centromere. (D) GENCODE genes with coordinates read from our own annotation. (E) A 385 kb TAD. (F) The NOTCH1-anchored loop.

6 Failure modes and controls

6.1 Shuffled-structure controls

ID	Control	Destroys	Rules out
S0	zero the signal	everything	establishes the sequence-only floor at identical parameter count
S1	permute across bins	alignment and autocorrelation	primary reliance probe
S2	circular shift, 10 Mb	alignment only	“any smooth auxiliary channel would do”
S3	distance-matched rewire	locus-specific structure	“structure is genomic distance in disguise”

Table 6: S3 is the control that can end the project.

Contact probability falls smoothly with genomic distance. If a model performs as well on distance-matched fake structure as on real structure, what it learned is a re-parameterised positional prior and the central claim is false however good the headline number looks.

The gate passes only if degradation under shuffling is positive with a 95% bootstrap confidence interval excluding zero *and* is at least twice the across-seed standard deviation of the real-structure runs. Significance alone is insufficient; the effect-size floor prevents a p -value standing in for an effect.

6.2 The eight failure modes

Failure	Symptom	Status
F1 bias absorbed into b_δ	trains to baseline loss	highest probability; mitigated by genome-wide standardisation
F2 softplus saturation	structural weights never move	observed ; b_g changed $-8 \rightarrow -4$
F3 structure inferable from sequence	structural input redundant	largely defused by Lee’s Evo 2 probe
F4 bin longer than memory horizon	trains to baseline loss	precondition confirmed ; mitigated by interpolation
F5 permeability gate stays at zero	half the mechanism inert	monitor the distribution of p
F6 gradient starvation	structural pathway never learns	1.7% of parameters behind two nonlinearities
F7 forward and reverse cancel	trains to baseline loss	two of eight features flip sign under RC
F8 structural encoder collapses	reduces to F1	eliminated if the encoder is removed

F1, F4 and F7 produce the same symptom: the model trains to baseline performance and raises no error. Four diagnostics therefore run continuously—the variance the structural pathway contributes to the timescale, the distribution of the permeability penalty, the growth of the

structural weight norm, and a probe asking whether the baseline model’s hidden states already predict the structural signal.

6.3 Memory horizon, measured

At the reference initialisation, horizons run from 0.6 tokens at the fast end to 994 at the slowest, median 14.9. A 5 kb bin is 5,000 tokens, so no channel-state pair spans a single bin; $\tau_{\max}/\text{bin} = 0.199$. Sixteen layers relaying extends reach to roughly 15,900 tokens, still short of a 385 kb TAD.

Δ is learned and can grow, so this is not proof of failure. It constrains the sequence-only baseline exactly as much as the structural model, which raises a question worth settling before spending GPU time: can a model of this size represent TAD-scale dependencies at all? **Phase 3 must therefore report the trained memory horizon, and that measurement gates Phase 4.**

7 Decisions of record

	Decision	Choice	Reasoning
1	Mechanism	structural bias in the recurrence	unreachable by contrastive alignment on output embeddings; contains hard state resets as its limit; cheap to falsify
2	Reverse complement	Caduceus-PH (augmentation)	equivariance is orthogonal to the hypothesis; coupling them makes a null result uninterpretable
3	Hi-C resolution	5 kb, 1 kb as ablation	comparability with CHROME; both levels are in the same file
4	Cross-species scope	out for this paper	Evo2HiC uses 177 species, HiC-Foundation 316; entering that lane splits compute against better-resourced groups
5	Permeability term	keep , via custom kernel	retains the soft-reset semantics and the link to future work

8 Change log

Every substantive change to the project is recorded here with its reason.

2026-08-07

- **Phase 0 completed.** Eleven papers mapped. Two competitors found that were not on the original reading list: Evo2HiC, which is closer to the thesis than CHROME, and Lee’s Evo 2 probe, which supports it.
- **Differentiator retired.** “Structure at training, sequence-only at inference” was dropped as a claim against Evo2HiC, whose DNA-only encoder has the same property. It still holds against CHROME.
- **Forward-citation sweep.** 318 papers checked; no work probes Akita or Orca representations on a non-folding task. The transfer gap stands.

- **Mechanism (a) selected;** (b) and (c) documented but not pursued.
- **Memory horizon measured.** $\tau_{\max} = 994$ tokens against a 5,000-token bin. Consequence: ϕ interpolation became mandatory, and a trained- τ measurement was added as a Phase 4 gate.
- **Pilot acquired and validated.** Insulation $r = +0.9969$ and compartments $r = +0.9759$ against independent 4DN tracks. A TAD and a ChIA-PET-corroborated loop confirmed visually.
- **Three assembly traps caught.** A 1,425 kb “loop” that was sparse-corner noise; an hg19 *ABL1* coordinate used to choose a GRCh38 region; and the standard GEO loop list for this dataset, which is hg19—detected because its largest chr9 anchor exceeds the length of GRCh38 chr9.
- **Dataset layer built.** Splits verified exact after a first version allowed held-out windows to straddle their region edge.
- **Model implemented; two bugs found by unit tests.** The reverse-complement sign flip was being applied to the encoded signal rather than to raw ϕ , where the encoder has already mixed all eight coordinates, making the operation meaningless. Separately, A was broadcast by trailing-axis alignment, which matches the state dimension against sequence length and only agrees when $L = d_{\text{inner}}$; the first test passed only because it used $L = 64$ with $d_{\text{inner}} = 64$.
- b_g **changed from -8 to -4 .** The gradient through softplus is $\sigma(b_g)$, so -8 attenuated the gate’s learning signal roughly 3000-fold—failure mode F2, self-inflicted by our own initialisation. Measured effect on the gate gradient: $2.06 \times 10^{-7} \rightarrow 7.69 \times 10^{-6}$.
- **Encoder bottleneck identified.** $\partial L / \partial s = W_{\Delta s}^T \partial L / \partial \Delta$ with $W_{\Delta s}$ zero-initialised, so the encoder receives exactly zero gradient on step zero and roughly 10^{-9} afterwards against 10^{-4} for the matrix in front of it. Recommendation: remove the encoder and feed the eight standardised features directly at $d_s = 8$, costing +1.700% parameters and eliminating F8.
- **Triton kernel written** for the permeability term, with a GPU validation harness. Unverified until executed.

9 Open items

- **Validate the Triton kernel on GPU** before any training run.
- **Confirm the encoder removal** ($d_s = 8$, no encoder), which changes parameter figures recorded in three documents.
- **Read CHROME line by line.** The summary here came from automated extraction of the PMC full text.
- **Obtain Orca’s Results section**, paywalled at *Nature Genetics* with no PubMed Central deposit. The claim that its representations were never evaluated on a non-folding task rests on the abstract, all ten extended-data captions and the section headings, not the body.
- **Re-check Evo2HiC** for a revision adding variant evaluation before Phase 4 commits GPU time; that would close the one contribution nothing currently contests.
- **Re-check Lee’s preprint** for peer-review status before it carries weight in a methods section.

10 What happens next

Phase 3 trains the sequence-only baseline, logging every hyperparameter, seed and metric before any structural model exists, so no flattering comparison can be selected after the fact. That run must also report the trained memory horizon, which decides whether a model of this size can

represent TAD-scale dependencies at all. If it cannot, the mechanism is re-scoped to sub-TAD structure or the model is enlarged, and that decision is taken before Phase 4 rather than after.

Every measured number in this document traces to a named script: `phase1_acquire.py`, `phase1_features.py`, `phase1_validate_visual.py`, `phase1_dataset.py`, `param_accounting.py`, `f4_memory_horizon.py`, `test_model.py`. Literature claims are sourced in `docs/related_work.md`; the mechanism is specified in `docs/architecture_spec.md`; data provenance is in `docs/data_card.md`.